LESSON PLAN

BASIC ELECTRICAL (23)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	22
Total No. of Lesson Plans	29	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Control system components
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Control system components
5. Specific Objectives	23.1.5.1 Trainee will be able to explain the construction and functioning of aircraft flight controls systems and its components.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Identification of control system components on aircraft model; Safety: Handle model parts gently; Supervision: Instructor monitors identification accuracy
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary flight controls of an aircraft?
(b) How does a pilot initiate a turn using the control yoke?
(c) Name the secondary flight controls found on modern aircraft.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Primary flight control surfaces • Control system linkage mechanisms • Hydraulic actuation systems • Electrical control components	L & Ds	PPT, IB	15 min
(b)	• Functioning of ailerons and elevators • Rudder operation and yaw control • Flap and spoiler mechanisms • Trim system operation	L & Ds	PPT, PDF	20 min
(c)	• Fail-safe mechanisms in controls • Redundancy in flight control systems • Maintenance checks for control links • Troubleshooting common control issues	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how hydraulic pressure assists in moving control surfaces.
(b) Describe the role of trim tabs in reducing pilot workload.
(c) Identify two fail-safe features in a typical flight control system.

12. Home Assignment and Follow Up	Prepare a brief note on the need for simple oxygen systems and the effects of lack of oxygen, as covered in LP 23.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Need for simple oxygen system and pressurized cabin effect of lack of o2, Types of 02 regulators. Fire detection and protection system:	Refer to the master syllabus reference list	LP 23

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min